AI-Driven Technical Change, the Real Interest Rate, and the Real Wage: Closed Forms in a Production Economy with Adjustment Costs

Research note (draft)

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Abstract

This note uses the production economy in Kiley (2000) (published as Kiley, 2004) as a blueprint for analyzing AI-driven technical change. That paper pairs Cobb-Douglas production with convex investment adjustment costs that are zero along the balanced growth path, so $Q = 1$ and market value, interest rates and factor prices have simple closed forms. I add a third input, AI capital, which combines with the traditional capital-labor bundle through a CES aggregator. AI technology improves either as disembodied progress (a factor Γ that augments the installed stock) or as embodied progress (a falling price P_A of new AI efficiency units, as in Greenwood, Hercowitz and Krusell, 1997). The balanced growth path has closed forms for the equilibrium real interest rate, the real wage, the labor share, Tobin's Q and the ratio of market value to output. With Cobb-Douglas aggregation the path is exact under permanent AI progress. With substitutes ($\varepsilon > 1$) the economy can tip into an AK regime, and with complements ($\varepsilon < 1$) there is a Baumol-type limit. Perfect-foresight transitions show that disembodied and embodied AI shocks with identical long-run effects have different paths for the real interest rate and the real wage.

1 The blueprint

Kiley (2000) studies a representative-agent economy (notation as in that paper). The household maximizes

$$U = E_t \sum_{j=t}^{\infty} B^{j-t} \frac{[C(j)^b L(j)^{1-b}]^{1-s}}{1-s}, \qquad L = 1 - N, \tag{2.1}$$

and holds claims to capital, $C(t) + Q(t)K(t+1) = (Q(t) + D(t))K(t) + W(t)N(t)$. Firms produce

$$Y = K(t)^a (Z(t)N(t))^{1-a} - \Psi[I(t)/K(t)]K(t), \qquad \Psi[I/K]K = 0.5\, c\, (I/K - x)^2 K, \quad x = g + d, \tag{2.7}$$

with $K(t+1) = I(t) + (1-d)K(t)$. The paper writes the adjustment-cost function as $C[\cdot]$; I write it as Ψ to avoid a clash with consumption. Optimality gives $1 + \Psi_I = Q$ (2.11) and the capital Euler equation (2.12). Because adjustment costs are zero when $I/K = g + d$, every balanced growth path has $Q = 1$. The steady-state results in the paper are then

$$r = -\log B + g \ (s = 1), \qquad \frac{aY}{K} = r + d, \qquad \frac{D}{Y} = \frac{a(r-g)}{r+d}, \qquad \frac{V}{Y} = \frac{a}{r+d}. \tag{3.2}$$

The calibration (Table 1 of the paper) is $s = 1$, $b = 0.25$, $a = 0.282$, $d = 0.066$, $r = 0.075$, with $c = 15$ for "moderate" adjustment costs.

The working paper's production function is Cobb-Douglas. The CES structure below is this note's extension. It enters through the AI aggregator, not the capital-labor bundle. The appendix of Kiley (2000) already introduces a second capital stock with its own investment price and adjustment-cost function, and argues that its accumulation tracks physical capital unless its relative price moves. Embodied AI progress is exactly such a movement in relative price.

2 The model with AI capital

Technology. The traditional bundle and gross output are

$$X = K^a(ZN)^{1-a}, \qquad F = \left[\omega^{1/\varepsilon} X^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} M^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad M = \Gamma A. \tag{1}$$

Here A is AI capital measured in efficiency units and M is effective AI. With $\varepsilon < 1$, AI complements the capital-labor bundle; with $\varepsilon > 1$, it substitutes for it. The distribution weights carry the exponent $1/\varepsilon$, so $1-\omega$ is the AI income share when the effective cost of AI is one, and the $\varepsilon \to 1$ limit is $F = X^{\omega} M^{1-\omega} / [\omega^{\omega}(1-\omega)^{1-\omega}]$.

Two kinds of AI progress. *Disembodied*: Γ grows at rate γ_Γ and raises the productivity of the entire installed stock. *Embodied*: new AI efficiency units cost P_A in output, and P_A falls at rate γ_q, with $P_A(t)/P_A(t+1) = 1 + \gamma_q$. This is the investment-specific technical change of Greenwood, Hercowitz and Krusell (1997): a unit of output buys $1/P_A$ efficiency units, but units already installed do not improve.

Accumulation and adjustment costs. Following (2.7) for both stocks,

$$K(t+1) = I_K(t) + (1-d_K)K(t), \qquad A(t+1) = I_A(t) + (1-d_A)A(t), \tag{2}$$

$$Y = F - 0.5c_K(I_K/K - x_K)^2 K - P_A\, 0.5c_A(I_A/A - x_A)^2 A, \tag{3}$$

and $C + I_K + P_A I_A = Y$. AI adjustment costs are scaled by P_A, so they stay in proportion to AI spending as P_A falls, which is the appendix specification with its own investment price. The parameters x_K, x_A equal steady-state investment rates.

Optimality. Labor demand is $W = F_X(1-a)X/N$. Investment gives $Q_K = 1 + c_K(I_K/K - x_K)$ and $Q_A = P_A[1 + c_A(I_A/A - x_A)]$. Write $m_{t+1} = B\,U_C(t+1)/U_C(t) = 1/(1+r_{t+1})$. The Euler equations are

$$Q_K(t) = m_{t+1}\big\{F_K(t+1) - \Psi_K(t+1) + \Psi_K'(t+1)\, i_K(t+1) + (1-d_K)Q_K(t+1)\big\}, \tag{4}$$

$$Q_A(t) = m_{t+1}\big\{\Gamma(t+1)F_M(t+1) + P_A(t+1)[\Psi_A'\, i_A - \Psi_A](t+1) + (1-d_A)Q_A(t+1)\big\}, \tag{5}$$

where $i = I/K$ or I/A and $\Psi_j(i) = 0.5c_j(i - x_j)^2$. The first is (2.12). The second is its AI analogue. The household conditions are $(1-b)C/(bL) = W$ and $U_C(t) = B\,E_t[U_C(t+1)(1+r(t+1))]$. Firm value is $V(t) = Q_K(t)K(t+1) + Q_A(t)A(t+1)$; by Hayashi (1982) this equals the present value of dividends $D = Y - WN - I_K - P_A I_A$.

3 Closed forms on the balanced growth path

Define the *effective cost of AI*, the rental price of one unit of M,

$$\lambda \equiv \frac{P_A\, u_A}{\Gamma}, \qquad u_A = (1+r)(1+\gamma_q) - (1-d_A) \;\approx\; r + d_A + \gamma_q. \tag{6}$$

The user cost u_A contains an obsolescence term γ_q: a unit bought today loses value as newer, cheaper units arrive. The user cost of traditional capital is $R_K = r + d_K$. Let $\theta \equiv 1 - b(1-s)$ be the inverse intertemporal elasticity (footnote 11 of the paper).

Proposition 1 (Balanced growth)**.** *Suppose either (i) Γ/P_A is constant, for any ε, or (ii) $\varepsilon = 1$ with Γ growing at γ_Γ and P_A falling at γ_q. On the balanced growth path with growth g, $Q_K = 1$, $Q_A = P_A$, hours are constant, and*

$$1 + r = (1+g)^\theta / B, \tag{7}$$

$$s_M = (1-\omega)\,\lambda^{1-\varepsilon} \quad \textit{(AI share of income)}, \tag{8}$$

$$\textit{labor share } \frac{WN}{Y} = (1-a)(1-s_M), \tag{9}$$

$$P_X = \big[(1-s_M)/\omega\big]^{\frac{1}{1-\varepsilon}} \;\; (\varepsilon \neq 1), \qquad P_X = \lambda^{-\frac{1-\omega}{\omega}} \;\; (\varepsilon = 1), \tag{10}$$

$$\frac{W}{Z} = (1-a)\Big[P_X\Big(\frac{a}{r+d_K}\Big)^a\Big]^{\frac{1}{1-a}}, \tag{11}$$

$$\frac{K}{Y} = \frac{a(1-s_M)}{r+d_K}, \qquad \frac{P_A A}{Y} = \frac{s_M}{u_A}, \tag{12}$$

$$\frac{V}{Y} = (1+g)\Big[\frac{K}{Y} + (1+\gamma_q)\frac{P_A A}{Y}\Big] \approx \frac{a(1-s_M)}{r+d_K} + \frac{s_M}{r+d_A+\gamma_q}, \tag{13}$$

$$N = \frac{b(1-a)(1-s_M)}{b(1-a)(1-s_M) + (1-b)\,C/Y}, \tag{14}$$

$$\frac{C}{Y} = 1 - (g+d_K)\frac{K}{Y} - \big[(1+g)(1+\gamma_q) - 1 + d_A\big]\frac{P_A A}{Y}. \tag{15}$$

Derivation. With Q's constant and adjustment costs zero, the Euler equations give $F_K = r + d_K$ and $\Gamma F_M = P_A u_A$, the latter because $Q_A = P_A$ falls at rate γ_q (a capital loss). Competitive pricing makes unit cost equal one, $1 = [\omega P_X^{1-\varepsilon} + (1-\omega)\lambda^{1-\varepsilon}]^{1/(1-\varepsilon)}$, where P_X is the shadow price of the bundle. This gives (8) and (10). The Cobb-Douglas bundle's cost function $P_X = (R_K/a)^a (W/[Z(1-a)])^{1-a}$, inverted, gives (11). The interest rate (7) is the household Euler equation with constant hours. Market value follows from $Q = 1$ and the timing of V.

Equation (13) is the AI generalization of (3.2): each capital stock is valued at its income share over its user cost. A compact way to read the wage formula is

$$\ln \frac{W}{Z} = \text{const} + \frac{\ln(1-s_M)}{(1-\varepsilon)(1-a)} - \frac{a}{1-a}\ln(r+d_K). \tag{16}$$

AI moves the real wage through two channels only: the share of income it absorbs (scaled by $1/(1-\varepsilon)$) and the interest rate. When $\varepsilon > 1$, a rising AI share raises the wage level even as it lowers the labor share. When $\varepsilon < 1$, cheaper AI shrinks its share and the wage gain is bounded.

Table 1: Cobb-Douglas balanced growth paths ($\varepsilon = 1$). Rates in percent. W/Z is relative to the no-progress path, with all three paths starting from the same Γ/P_A at $t = 0$.

	g	r	u_A	W/Z	WN/Y	K/Y	P_AA/Y	V/Y
No AI progress	1.50	7.50	27.5	1.000	0.696	1.94	0.109	2.08
Disembodied, $\gamma_\Gamma = 0.10$	1.92	7.94	27.9	0.987	0.696	1.88	0.107	2.03
Embodied, $\gamma_q = 0.10$	1.92	7.94	38.7	0.973	0.696	1.88	0.077	2.00

3.1 Cobb-Douglas: an exact balanced growth path with permanent AI progress

Proposition 2 ($\varepsilon = 1$)**.** *With Γ growing at γ_Γ and P_A falling at γ_q,*

$$\ln(1+g) = \ln(1+g_Z) + \frac{1-\omega}{(1-a)\,\omega}\big[\ln(1+\gamma_\Gamma) + \ln(1+\gamma_q)\big], \tag{17}$$

AI capital in efficiency units grows at $(1+g)(1+\gamma_q)$, $s_M = 1-\omega$, the labor share is $(1-a)\omega$, and r follows from (7). The wage grows at g, and its level relative to Z is $W/Z = (1-a)\big[\lambda^{-\frac{1-\omega}{\omega}}(a/R_K)^a\big]^{\frac{1}{1-a}}$, with $\lambda = P_Au_A/\Gamma$ falling at $(1+\gamma_\Gamma)(1+\gamma_q)$.

Disembodied and embodied progress at the same rate deliver the same growth rate and the same real interest rate. They differ in levels, because embodied progress raises the AI user cost by the obsolescence term. Relative to disembodied progress at the same rate, the wage level is lower by $\frac{1-\omega}{(1-a)\omega}\ln[u_A^{\text{emb}}/u_A^{\text{dis}}]$ log points, P_AA/Y is lower by the factor $u_A^{\text{dis}}/u_A^{\text{emb}}$, and so is the AI contribution to V/Y. Table 1 shows this.

As in Kiley (2000), faster growth raises r one-for-one with $\theta = 1$ and *lowers* K/Y and V/Y. AI-driven growth does not mechanically raise valuation ratios. Embodied progress lowers V/Y further because AI assets depreciate economically as their price falls.

3.2 Substitutes ($\varepsilon > 1$): the AK regime

Proposition 3 ($\varepsilon > 1$)**.** *Let $\bar{A} \equiv (1-\omega)^{1/(\varepsilon-1)}$ and $r_Z = (1+g_Z)^\theta/B - 1$. With Γ/P_A constant, an interior balanced growth path exists if and only if $\lambda(r_Z) = P_A(r_Z + d_A)/\Gamma > \bar{A}$, i.e. $s_M < 1$. Otherwise the economy converges to an AK path in which $F \to \bar{A}\,\Gamma A$ and*

$$r = \bar{A}\,\Gamma/P_A - d_A, \qquad 1+g = [B(1+r)]^{1/\theta} > 1+g_Z. \tag{18}$$

On this path the bundle's shadow price grows at $\ln G_P = \frac{(1-a)(\ln(1+g)-\ln(1+g_Z))}{\varepsilon(1-a)+a}$, so

$$\Delta\ln W = \ln(1+g_Z) + \frac{\ln(1+g)-\ln(1+g_Z)}{\varepsilon(1-a)+a}, \qquad \Delta\ln(\textit{labor share}) = -(\varepsilon-1)\ln G_P. \tag{19}$$

In the AK regime, the real wage grows faster than Z but more slowly than output, so the labor share declines toward zero while workers become richer. The interest rate is pinned by the AI rental rate, not by g_Z. If Γ/P_A keeps rising, so does the AK productivity $\bar{A}\Gamma/P_A$, and r and g rise without bound. This is the explosive-growth knife edge. At the baseline calibration with $\varepsilon = 1.5$, the threshold is reached only after Γ/P_A rises about 1,100-fold from its initial level, so the 20-fold shock studied below stays in the interior regime.

Table 2: 20-fold fall in the effective cost of AI: closed-form new BGP and transition statistics.

	s_M new BGP	WN/Y new BGP	W/Z new/old	V/Y new BGP	peak r (pct.) disemb.	emb.	W/Z at $t=10$ disemb./emb.
$\varepsilon = 0.5$	0.007	0.713	1.068	2.04	8.18	7.93	1.039 / 1.014
$\varepsilon = 1$	0.030	0.696	1.138	2.08	8.59	8.22	1.048 / 1.015
$\varepsilon = 1.5$	0.134	0.622	1.372	2.25	9.66	8.93	1.061 / 1.015

3.3 Complements ($\varepsilon < 1$): the Baumol limit

Proposition 4 ($\varepsilon < 1$)**.** *If $\Gamma/P_A \to \infty$, then $\lambda \to 0$, $s_M = (1-\omega)\lambda^{1-\varepsilon} \to 0$, $g \to g_Z$, $r \to r_Z$, the labor share $\to 1-a$, $P_X \to \omega^{-1/(1-\varepsilon)}$, and*

$$\frac{W_\infty}{W_0} = (1 - s_{M,0})^{-\frac{1}{(1-\varepsilon)(1-a)}}. \tag{20}$$

AI becomes so cheap that it stops being a bottleneck. Output is limited by the traditional bundle, and the total wage gain from AI is capped by its initial income share. With $s_{M,0} = 0.03$, $\varepsilon = 0.5$ and $a = 0.282$, the cap is 8.9 percent.

4 Transitions: embodied versus disembodied AI shocks

The script `ai_model.py` solves the perfect-foresight path with a Newton method on the stacked equilibrium conditions, with all variables detrended by Z. The calibration is the paper's: $a = 0.282$, $b = 0.25$, $s = 1$, $d_K = 0.066$, $c_K = 15$, initial $r = 0.075$. Trend growth is $g_Z = 0.015$, where the paper used 0 for convenience. The AI block has $d_A = 0.20$, $c_A = 15$, an initial AI income share of 3 percent, and Γ_0 normalized so that $\lambda_0 = 1$, which makes the initial path identical for all ε. At $t = 0$ agents learn that Γ/P_A will rise 20-fold along a logistic S-curve centered on year 10. The rise comes either through Γ (disembodied) or through P_A (embodied). Both shocks have the same initial and final balanced growth paths, which Proposition 1 gives in closed form, so all of the difference lies in the transition.

Results.

1. *The real interest rate rises temporarily in both cases, and more under disembodied progress.* Disembodied progress raises the productivity of AI already in place, so output and expected consumption growth rise as the S-curve unfolds, and r rises with them. The peak is 8.6 percent in the Cobb-Douglas case and 9.7 percent with $\varepsilon = 1.5$. Embodied progress pays off only through new vintages, and accumulation is slowed by adjustment costs. The rise in r is therefore smaller (8.2 and 8.9 percent at the peak), comes later and lasts longer. Under complements the rise is smallest, because AI's share collapses as it becomes cheap. In every case r returns to r_Z: a one-time level shift has no long-run effect on r. Only a permanent change in the growth rate does (Table 1).

2. *The real wage rises in all cases, by an amount set by ε.* The long-run gain is 6.8 percent with complements (close to the 8.9 percent Baumol cap), 13.8 percent with Cobb-Douglas, and 37 percent with substitutes, all from (16). Under embodied progress the wage lags by

roughly five years: ten years in, the wage is up 1.5 percent under embodied progress versus 4–6 percent under disembodied progress.

3. *Labor share.* The gross labor share moves only through s_M. It rises with complements and falls with substitutes, from 69.7 to 62.2 percent at $\varepsilon = 1.5$. In the Cobb-Douglas case it is constant gross of adjustment costs. The temporary rise under embodied progress reflects the adjustment costs of the AI investment boom, which reduce net output.

4. *Valuation.* Embodied progress generates an AI investment boom with $Q_A/P_A > 1$ and a temporary rise in V/Y. Disembodied progress in the complements case *lowers* V/Y in the transition, as in the paper's growth experiment. In the long run, V/Y follows (13) and rises only when the AI share rises ($\varepsilon > 1$).

5 Verification

The script checks every closed form against the equilibrium conditions directly. It builds level paths, without detrending, from the closed forms and evaluates the production function, both Euler equations, the labor condition and the household Euler equation. It then compares V with the present value of dividends. It does this for the Cobb-Douglas path with permanent embodied and/or disembodied progress and for CES steady states with $\varepsilon \in \{0.5, 1.5\}$. It also checks the Baumol limit, the consistency of the AK threshold, and that every transition converges to the closed-form new path. Table 3 reports the largest error for each check.

Table 3: Verification: maximum absolute (or relative) errors.

Check	max abs. error
CD BGP: F = Y (prod. fn. at closed form)	6.2e-15
CD BGP: labor FOC (1-b)C/(bL) = W	1.6e-15
CD BGP: Euler K (2.12)	9.8e-15
CD BGP: Euler A	1.0e-14
CD BGP: r = (1+g)$^t heta/B - 1$	2.2e-15
CD BGP: wage W/Z closed form	3.3e-15
CD BGP: labor share	2.2e-16
CD BGP: V/Y closed form	1.6e-14
CD BGP: V = PV(dividends) (Hayashi)	1.9e-14
CES steady state: F = Y (prod. fn. at closed form)	8.9e-16
CES steady state: labor FOC (1-b)C/(bL) = W	1.0e-15
CES steady state: Euler K (2.12)	1.2e-14
CES steady state: Euler A	1.0e-14
CES steady state: r = (1+g)$^t heta/B - 1$	2.2e-15
CES steady state: wage W/Z closed form	8.9e-16
CES steady state: labor share	4.4e-16
CES steady state: V/Y closed form	1.2e-14
CES steady state: V = PV(dividends) (Hayashi)	1.2e-14
Baumol limit W/Z (e=0.5)	8.4e-08
AK threshold consistent (e=1.5)	0.0e+00
AK threshold: g AK - g Z at threshold	9.7e-17
transition end W/Z vs closed form, e=0.5, disembodied	2.0e-05
transition end W/Z vs closed form, e=0.5, embodied	3.2e-05
transition end W/Z vs closed form, e=1.0, disembodied	5.3e-05
transition end W/Z vs closed form, e=1.0, embodied	8.8e-05
transition end W/Z vs closed form, e=1.5, disembodied	3.1e-04
transition end W/Z vs closed form, e=1.5, embodied	4.9e-04
transition Newton residual (max)	3.7e-12

6 Extensions

- Calibrating ε, d_A and γ_q to AI and software investment data, and comparing the implied paths for r with long-term real yields.
- Letting AI substitute for labor within the bundle (nested CES with task structure) to separate wage-level from labor-share effects.
- Adding an equity premium as in Section 3 of the paper, by moving B, to separate growth-driven from discount-rate-driven valuation.

References

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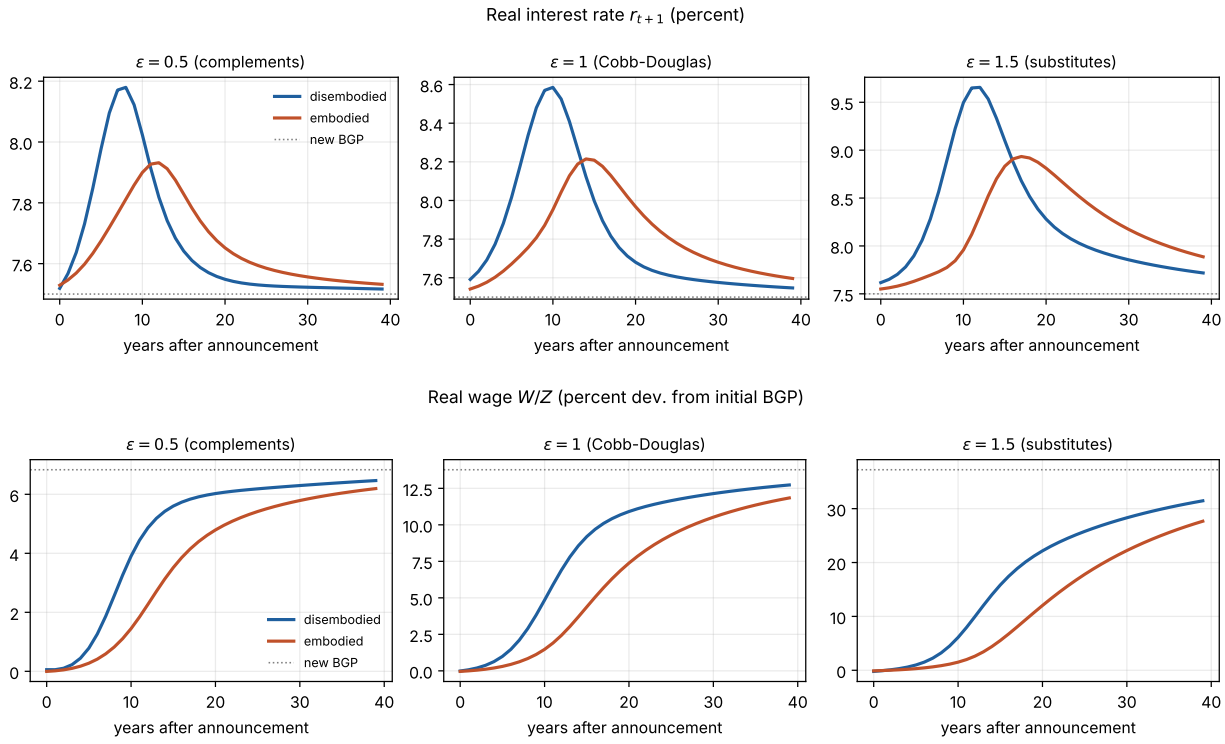


Figure 1: Real interest rate and real wage after an S-curve AI shock. Dotted lines: closed-form new BGP.

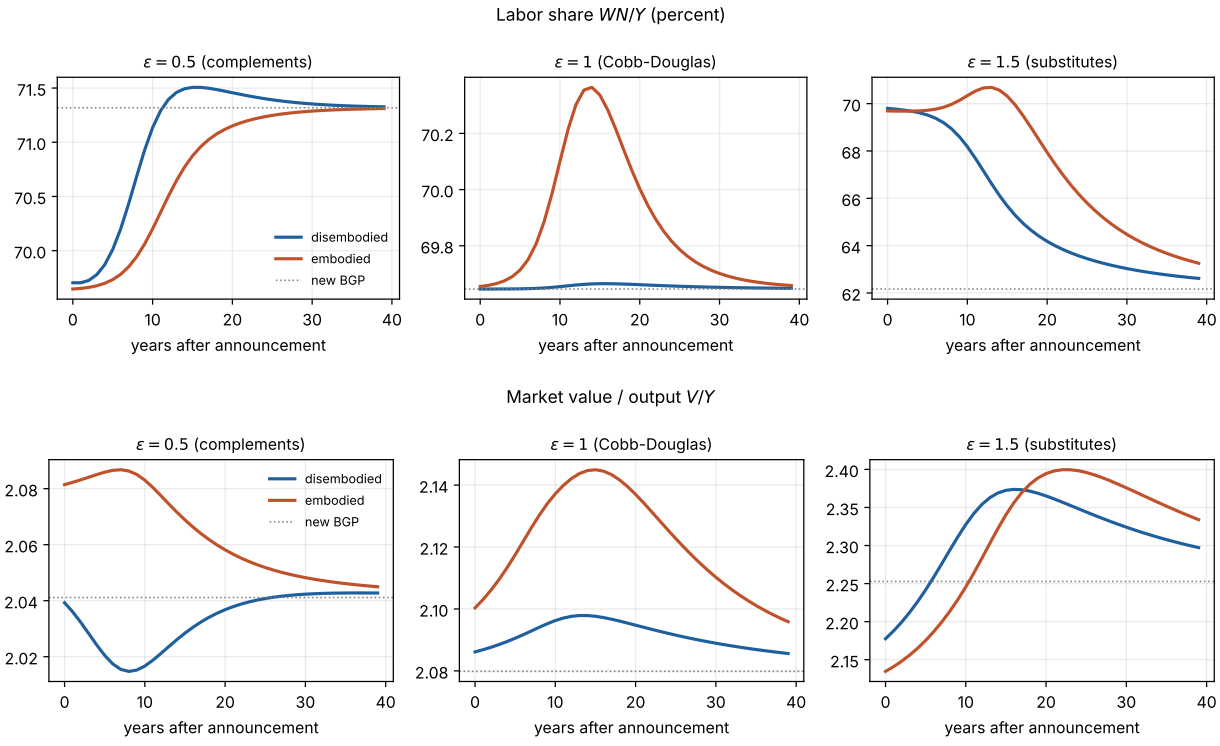


Figure 2: Labor share (relative to output net of adjustment costs) and market value relative to output.